MATH 350-2 Advanced Calculus

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Outline

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1. Real Analysis Lecture 11
   - More on metric spaces
   - Metric subspaces
   - Limits of sequences

Open sets

Definition

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The set A is **open** if every point in A is an interior point, or equivalently $\text{int}(A) = A$.

Challenge!

Problem

Consider the metric space $(\mathbb{R}, d_{\text{disc}})$ where d_{disc} is the discrete metric.
What sets are open sets?

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A point $x \in M$ is an **accumulation point** if for all $r > 0$ the ball $B_M(x; r)$ contains an element of A different from x.

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A point $x \in M$ is an **accumulation point** if for all $r > 0$ the ball $B_M(x; r)$ contains an element of A different from x.
The set $\overline{A}$ of all adherent points of A is called the **closure** of A.

Challenge!

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Prove the following (Apostol Theorem 3.36). If (M, d) is a metric space, $U \subseteq M$ is open, and $C \subseteq M$ is closed, then $U \setminus C$ is open and $C \setminus U$ is closed.

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Theorem (Apostol Theorem 3.37)

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(d) $A = \overline{A}$

Unions and intersections

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Open balls in S:

$$B_S(x; r) = B_M(x; r) \cap S.$$

Example

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Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

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Furthermore $U \cap S = V$. □

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We need to show that $V := U \cap S$ is open.

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Since $x \in V$ was arbitrary, this shows that V is open in S. $\square$

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We say $\{x_n\}$ **converges** to L and L is the **limit of the sequence.**

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Since $\epsilon > 0$ was arbitrary, this proves $\{x_n\}$ converges to 1. $\square$

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With the discrete metric $d(x_n, L) < 1$ implies $x_n = L$.
This means $x_n = 1/n = L$ for all $n \geq N$, which is a contradiction! □

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